# Chloé Weckel

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## Current position

2024-present **PhD on *Spatiotemporal modelling of signalling pathways: impact of endosomal compartmentalisation and application to gonadotropin receptors***, *team Musca INRAE-Inria-CNRS, Palaiseau / team BIOS, INRAE, Centre Val de Loire, Tours*, France.

- Developed ODE models of compartmentalized cAMP signalling based on chemical reaction networks, analysing how trafficking parameters shape signal dynamics and drive bistability
- Calibrated models to experimental BRET data (cAMP biosensors) using parameter estimation and model selection
- Designed and analysed hybrid stochastic models (piecewise-deterministic Markov processes) combining continuous dynamics with discrete random events

***Keywords***: ODE and PDMP models, parameter estimation, GPCR signaling, gonadotropin receptors.

Advisors: Romain Yvinec (team BIOS, INRAE/team Musca, Inria), Stefan Haar (team Musca, Inria) and Frédéric Jean-Alphonse (team BIOS, INRAE/ team Musca, Inria).

## Publications

2026 **Weckel C, Gourdon J, Darrigade L, Jugnarain V, Crépieux P, Reiter E, et al.**, Spatiotemporal modeling of GPCR signaling: The role of endosomal dynamics and receptor recycling. PLoS Comput Biol 22(9): e1014790.
https://doi.org/10.1371/journal.pcbi.1014790

2026 **Raynaud P, Gourdon J, Jugnarain V, Jean-Alphonse F, Berthet L, Moreau V, Weckel C, Vaugrente O, Gauthier C, Vallet A, Yvinec R, Anujan P, Naquin D, Boulo T, Pichon J, Gauthier C, Klett D, Combarnous Y, Sibille N, Hanyaloglu A, Reiter E, Bruneau G and Crépieux P**, A single domain intrabody biases the follicle-stimulating hormone receptor subcellular trafficking. Commun Biol.
https://doi.org/10.1038/s42003-026-11015-z

## Preprint

2026 **Darrigade L, Weckel C, et Yvinec R**, A stochastic model of compartmentalized intracellular signalling.
⟨hal-05794502⟩

## Internships

February-June 2023 **M2-internship *Spatio-temporal dynamics of signaling networks***, *team BIOS, INRAE Centre Val de Loire*, Tours, France.

***Keywords***: chemical reaction networks, ODE models, compartmentalization, signaling, endosomes, PDMP model.

Advisor: Romain Yvinec (team BIOS, INRAE/team Musca, Inria)

January-May 2022 **M1-internship on *Bienaymé-Galton-Watson process applied to an epidemiological context, in particular Covid-19***, *MAP5, University of Paris Cité*, Paris, France.

***Keywords***: branching process, epidemiology.

Advisor: Amandine Weber (MAP5, University of Paris Cité).

May-June 2021 **L3-internship on *Mathematical models of CD8 T immune response***, *MAP5, University of Paris Cité*, Paris, France.
***Keywords***: ODEs model, immunology.

Advisor: Fabien Crauste (MAP5, University of Paris Cité).

## Education

2021-2023 **Master Mathematics, Modeling and Learning**, *University of Paris Cité*, Paris.

2018-2021 **Bachelor's degree in Mathematics**, *University of Paris Cité*, Paris.

2018-2021 **Bachelor's degree in Biomedical Sciences**, *University of Paris Cité*, Paris.

## Position of responsibility

2025-present **Organiser of Musca Seminars**, *Inria, Paris-Saclay*.

**Co-organiser of PhD days of the PRC unit (Physiologie de la Reproduction et des Comportements)**, *INRAE, Centre Val de Loire, Tours*.

## Oral communications

**2026**.
- JMBS (Maths, Biology and Health Days) (Journées Maths-Bio-Santé), Lille, France. Travel grant from "the Réseau Thématique Maths-Bio-Santé"
- ECMTB, Graz, Austria. Travel grant from ESMTB
- IDP days, Orléans, France
- Bonn Conference on Mathematical Life Sciences, Bonn, Germany. Travel grant from the conference
- GT BIOSS, online

**2025**.
- JBS (Biostatistics Days) (Journées Biostatistiques), Montpellier, France. Travel grant from "the Réseau Thématique Maths-Bio-Santé"
- GPCR forum virtual conference, online
- CompsysBio, Aussois, France. Travel grant from the conference
- TOP days (Tours, Orléans, Poitiers), Tours, France

**2024**.
- Forum for Young Mathematicians, Montpellier

## Posters

**2026**.
- DIGIT-BIO, Montpellier

**2025**.
- JMBS (Maths, Biology and Health Days) (Journées Maths-Bio-Santé), Montpellier, France. Travel grants obtained from "the Réseau Thématique Maths-Bio-Santé"
- i-GPCR Net, Barcelona, Spain

## Teaching

**2026**.
- L1 Computer Science "Calculus et Raisonnement", 30h, University of Tours, France
- L1 Mathematics Analysis, 36h, University of Tours
- L2 Biology Statistics, 18h, University of Tours, France
- L1 Biology Mathematical tools, 6h, University of Tours, France

**2025**.
- L2 Biology Statistics, 18h, University of Tours, France
- L1 Biology Mathematical tools, 6h, University of Tours, France

## Science communication

2026 **Girls and Maths**, *Participation in the organization of the day with high school students*, University of Tours, France.

2025-2026 **Chiche**, *Interventions in high school classes to introduce the world of digital and mathematical research (8h)*, France.

2025 **Scientific mediation mission**, *"Les Sciences c'est leur chance" (32h) in 2 primary school classes*, University of Tours, France.

## Computer skills

Programming Languages: Python, R

## Awards

2026 **Special mention for the prize "La Main à la pâte"**, *in the Science Acamedic for the scientific mediation mission "Les Sciences c'est leur chance"*.

2022-2023 **Scholarship programme**, *Recipient of the scholarship programme PGSM Master (Paris Graduate School of Mathematical Sciences).* .

## Professional and volunteer experience

2023-2024 **Sailing trip**, *from France to Senegal followed by volunteer work (support in schools, projects tracking) with associations in Senegal*.

Since 2019 **Co-founder and treasurer of the association**, *"Les Loustics sous le Baobab" in support of the Senegalese association "Enfant d'abord" (education, health, awareness and self-fulfillment)*.

2019-2022 **Volunteering**, *for several months with the "Enfant c'abord" association in Senegal*.

2019-2024 **Sailing instructor**, *two months each summer for adults and children*.

## Referees

**Dr. Romain Yvinec**
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**Dr. Stefan Haar**
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